Practical 3 guidelines

- This form will not count any marks, only the answers you submit to ClickUP will count marks.

- Use the decimal point for number separation. Do not use the comma. eg. **Use 2.1** instead of 2,1. Marks will be lost if the comma is used. Also, **do not supply units!!**

- **Supply answers correct to 2 decimal places.**
- Handle your Op-Amp as little as possible. They are fragile and very sensitive.

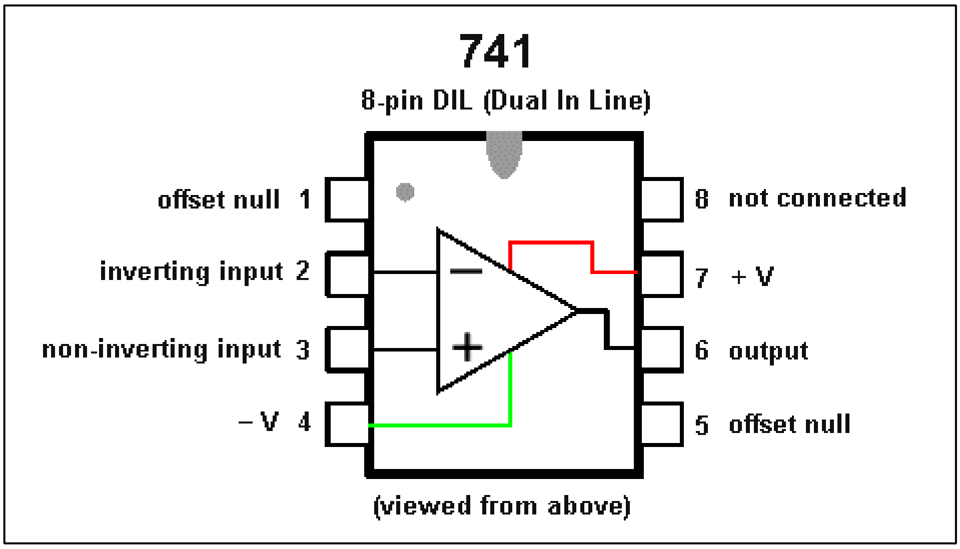


Figure 1: Op-amp pin layout.

Notice the **<u>dot</u>** in the one corner of the op-amp; this indicates where pin 1 is.

- There are **two (2) questions** in this practical, both deal with **Op-amps**

Example Op-Amp circuit set-up

The figure below is a simple op-amp circuit which should give you an idea of where to- and how to place the op-amp in your circuit.

Important: THIS IS NOT A CIRCUIT THAT YOU HAVE TO BUILD!

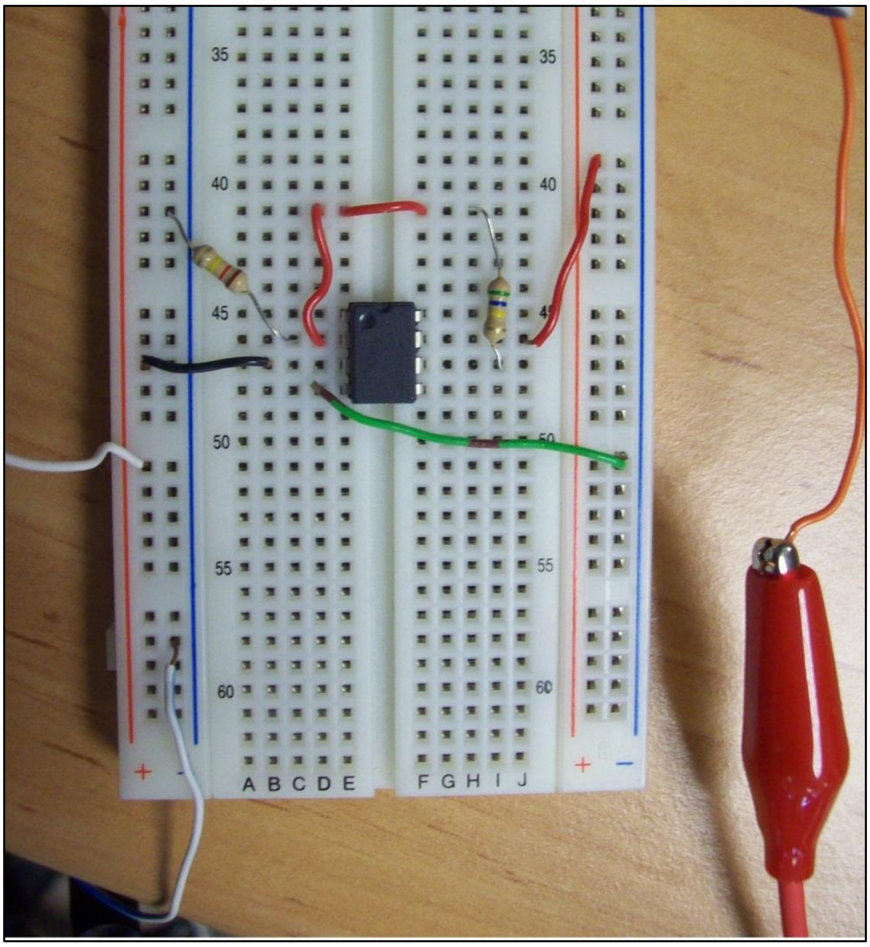


Figure 2: Example op-amp circuit

Setting up the power supply

Before you build the op-amp circuit. First set up the power supply correctly by performing the following steps. **DO THIS BEFORE YOU START WITH THE PRACTICAL.**

1. Turn the "Tracking ratio" dial on the power supply to the right and set it to "Fixed". In Figure 3, the "Tracking Ratio" it is indicated as label ***A***. This will enable you to set both the +20 V (label ***B***) and -20 V rails (label ***C***) of the power supply at the same time to the same value, but one rail will be positive and the other negative.

2. Set the voltage to 12 V (label ***D***).

3. Press the +20 V button (label ***E***) and check if the +20 V rail (label ***B***) is correctly supplying +12 V by using the multimeter. Then press -20 V (label ***F***) and check if the -20V rail (label ***C***) is supplying -12 V by using the multimeter.

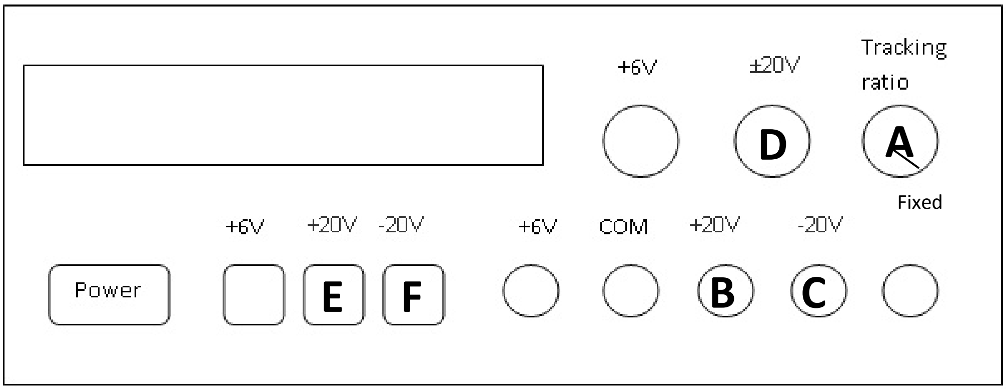


Figure 3: Ensuring the power supply is working correctly.

After you have set this up, take note of the following figure. This is what you will use during the practical.

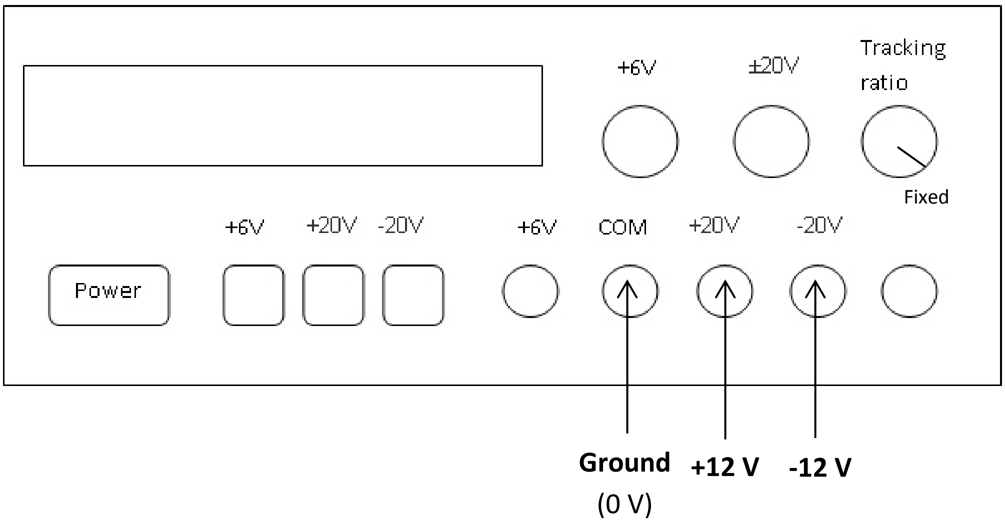


Figure 4: How the power supply will be used in the practical.

Question 1

The circuit diagram in Figure 5 illustrates the op-amp circuit that you will build. Before building the circuit, answer the following theoretical question.

Let the output voltage be V_o and the input signal V_{in}. Perform theoretical analysis on the circuit in Figure 5 and set up the equation in the following form:

$$V_0 = (A)V_{in}$$

Supply the answer for ***A***:

1.1) A= _______

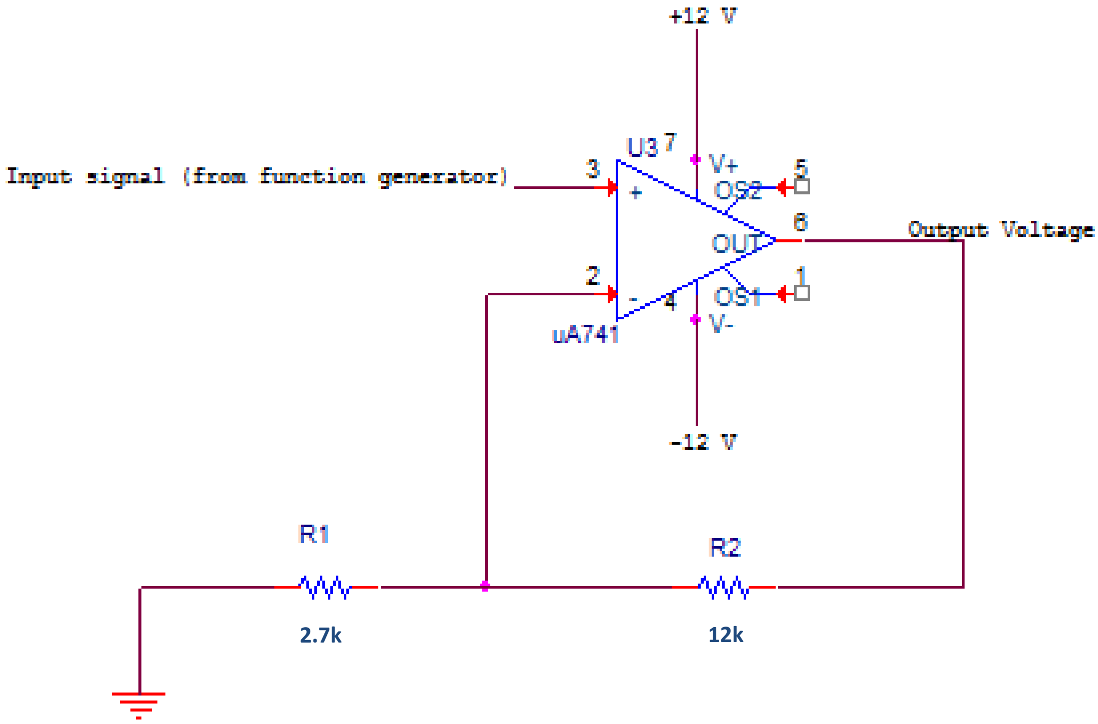


Figure 5. The circuit that you will build for Question 1.

Now build the circuit by performing the following steps. Make sure you have set up the power supply correctly as described in the previous section.

a) Place the op-amp in the middle of the breadboard (as in Figure 2).
b) Connect pin 6 of the op-amp to pin 2 of the op-amp with the **12 kΩ** resistor.
c) Connect pin 2 of the op-amp to ground (0 V) of the power supply with a **2.7 kΩ** resistor. Refer to Figure 4 if you are unsure about the ground.

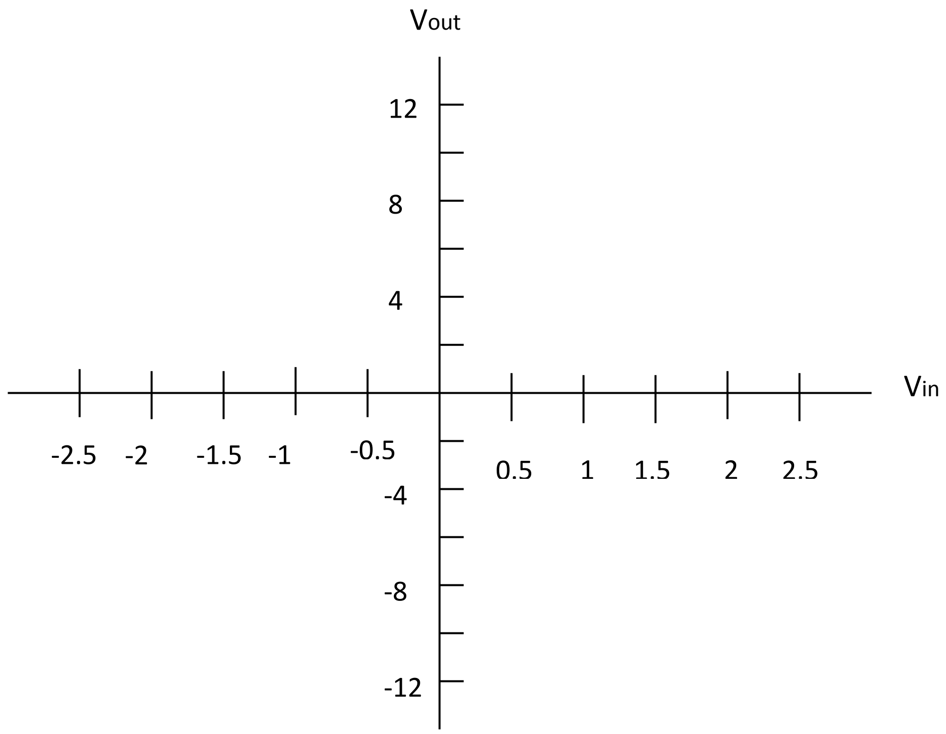
   HINT: Connect the ground from the power supply to a spot on your breadboard. All the connections that must go to ground can then merely be connected to that specific part on your breadboard

d) Connect pin 4 of the op-amp to the - 12 V rail of the power supply. Refer to Figure 4.
e) Connect pin 7 of the op-amp to the + 12 V rail of the power supply. Refer to Figure 4.
f) Connect pin 3 of the op-amp to the **function generator(gen out of the oscilloscope).** Make sure you connect pin 3 to the red part of the cable coming from the function generator. Connect the black part to the ground of the power supply.
g) Connect pin 6 of the op-amp to the oscilloscope's channel 1. This is where your output will be displayed (again, the black part of the cable should go to ground).
h) Set up the function generator by pushing the **Wave Gen** button on the oscilloscope. Under **waveform**, select **DC**.
i) Select the **Meas** button. Add the following measurement: Source: 1, **Type: Maximum**.
j) Press the **Wave** Gen button again. Here, the **Offset** button will be used to set the input voltages in the next step.
k) Complete the table below by setting the input voltages to the values in the table and then reading off the **Max** measurements at the different input voltages from the oscilloscope.

Question Number	**Input Voltage (DC offset) In Volt**	**Output Voltage (Max) in Volt**
1.2	-2.5	
1.3	-2	
1.4	-1.5	
1.5	-1	
1.6	-0.5	
1.7	0.5	
1.8	1	
1.9	1.5	
1.10	2	
1.11	2.5	

Table 1. Results from question 1.

Use the values from Table 1 to draw the graph of V_{out} versus V_{in}



Calculate the gradient of the straight line (linear region) of the op amp.

1.2) Gradient =

What is the percentage difference between the gradient and the theoretical gain of the op-amp (A) calculated in (1).

Hint: : $\%\text{difference} = \frac{|A-Gradient|}{A} \times 100$

1.3) %Difference =

1.4) The circuit is an example of a ________amplifier (inverting/non-inverting)

Question 2

The circuit diagram in Figure 6 illustrates the circuit you will build for Question 2. Before building the circuit, answer the following theoretical questions.

Let the output voltage be ***V*** V_o and the input signal be V_{in}. Perform theoretical analysis and set up the equation in the following form:

$$V_0 = (A)V_{in}$$

Supply the answer for ***A***:

2.1) A=

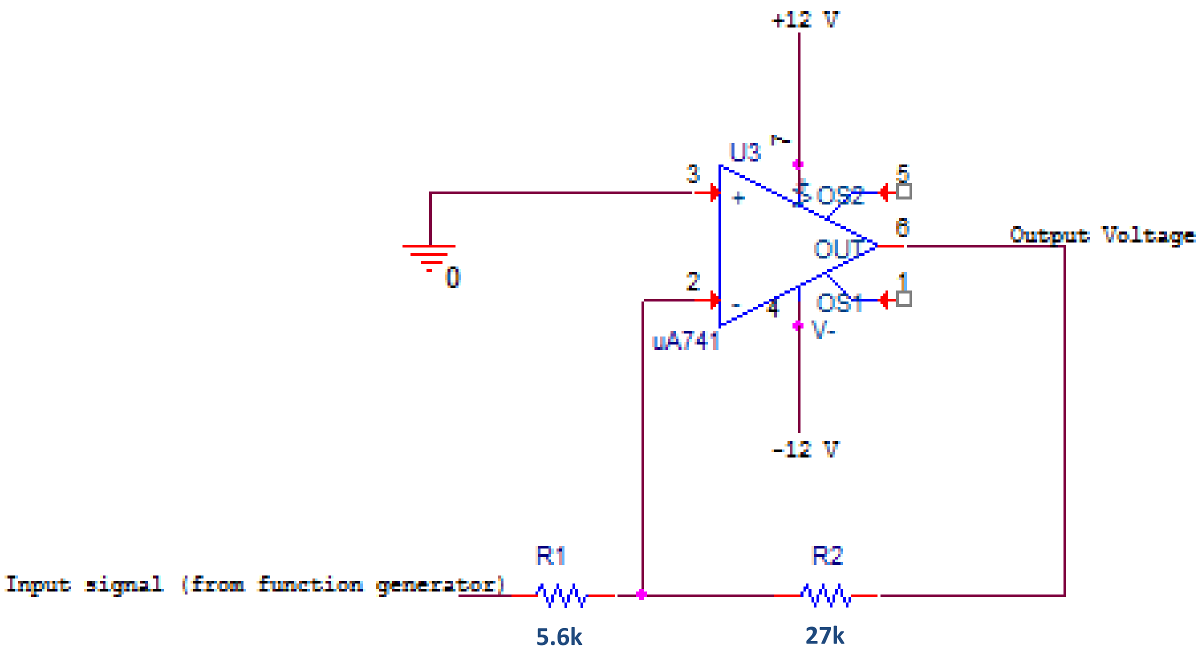


Figure 6. The circuit that you will build for Question 2.

Now build the circuit by performing the following steps. Make sure you have set up the power supply correctly as described earlier.

a) Place the op-amp in the middle of the breadboard (as in Figure 2).
b) Connect pin 6 of the op-amp to pin to pin 2 of the op-amp with the **27 kΩ** resistor.
c) Connect pin 3 of the op-amp to ground (0 V) of the power supply. Refer to Figure 4.

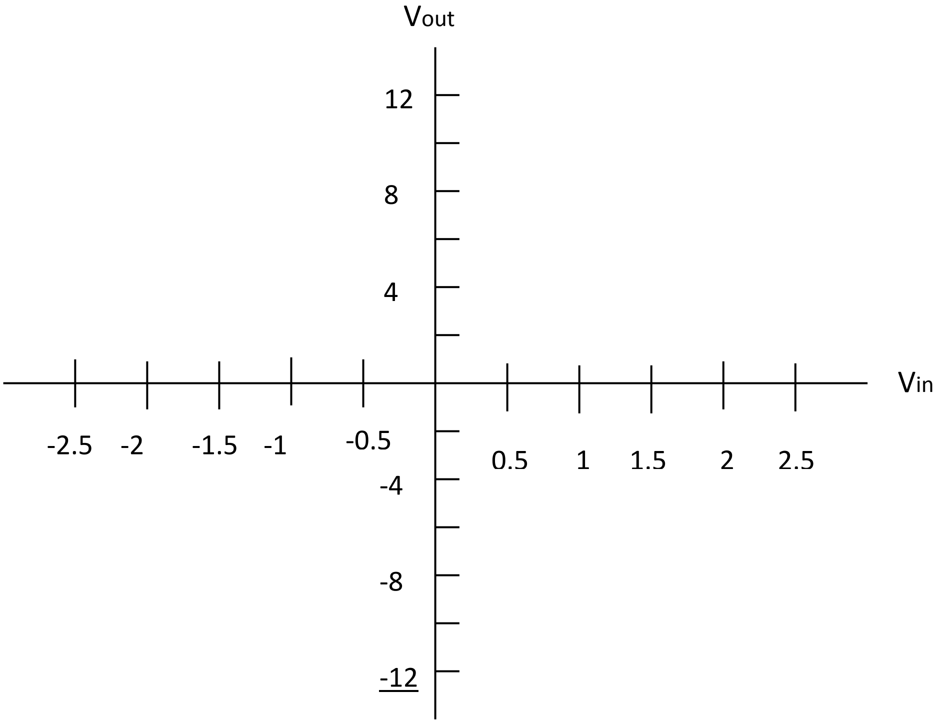
   HINT: Connect the ground from the power supply to a spot on your breadboard. All the connections that must go to ground can then merely be connected to that specific part on your breadboard

d) Connect pin 4 of the op-amp to the - 12 V rail of the power supply. Refer to Figure 4.
e) Connect pin 7 of the op-amp to the + 12 V rail of the power supply. Refer to Figure 4.
f) Connect pin 2 of the op-amp to the **function generator(gen out** of the oscilloscope**)** through the **5.6 kΩ** resistor . Make sure you connect it to the red part of the cable coming from the function generator. Connect the black part to ground of the power supply.
g) Connect pin 6 of the op-amp to the oscilloscope's channel 1. This is where your output will be displayed. Remember to connect the black part of the oscilloscope cable to ground.
l) Set up the function generator by pushing the **Wave Gen** button on the oscilloscope. Under **waveform**, select **DC**.
m) Select the **Meas** button. Add the following measurement: Source: 1, **Type: Maximum**.
n) Press the **Wave** Gen button again. Here, the **Offset** button will be used to set the input voltages in the next step.
o) Complete the table below by setting the input voltages to the values in the table and then reading off the **Max** measurements at the different input voltages from the oscilloscope.

Question Number	**Input Voltage (DC offset) In Volt**	**Output Voltage (Max) in Volt**
2.2	-2.5	
2.3	-2	
2.4	-1.5	
2.5	-1	
2.6	-0.5	
2.7	0.5	
2.8	1	
2.9	1.5	
2.10	2	
2.11	2.5	

Table 2. Results from question 2.

Use the values from Table 2 to draw the graph of V_{out} versus V_{in}



Calculate the gradient of the straight line (linear region) of the op amp.

2.2) Gradient =

What is the percentage difference between the gradient and the theoretical gain of the op-amp (A) calculated in (1).

Hint: : $\%\text{difference} = \frac{|A - Gradient|}{A} \times 100$

2.3) %Difference =

2.4) The circuit is an example of a _________ amplifier (inverting/non-inverting)

Question 3

Instructions

Build the circuit in Figure 7 and use the function generator to generate a sine wave with frequency 15kHz, 2 Vpp Amplitude and 2 V offset that will be your Vin. Connect you oscilloscope over the R3 resistor and then complete Table 3. **Each student will have a different value for R1 and must build the circuit with the value for R1 generated on clickUP.**

R1 = Obtain from ClickUP
R2 = 10kΩ
R3 = 27kΩ
R4 = 100kΩ

NB! The value for R1 is randomly chosen on ClickUP.

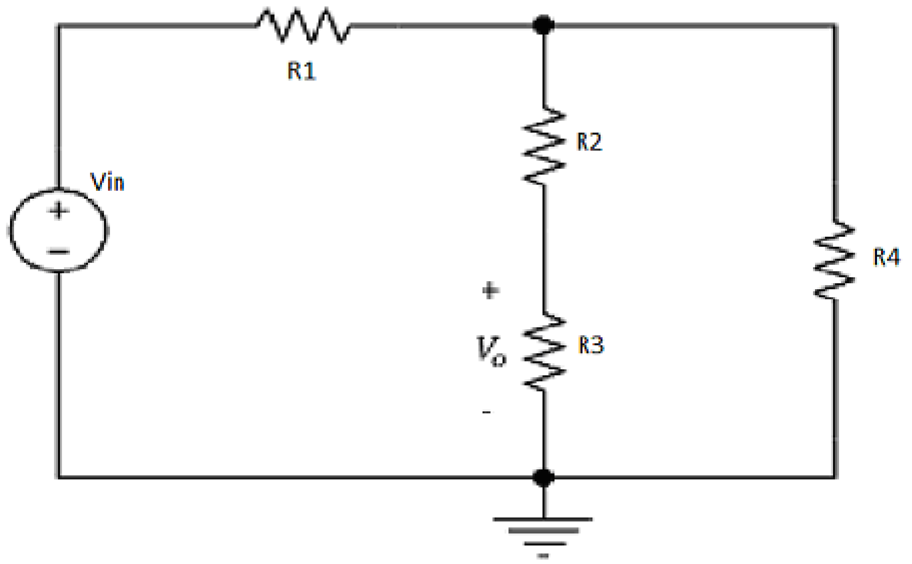


Figure 7. Circuit diagram for question 3

Amplitude [V]	Measured Frequency [kHz]

Table 3. Measured values for the circuit in Figure 7.

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